

TELECOM CHURN PREDICTION PROJECT

PROBLEM STATEMENT:

Customer churn poses a significant threat to profitability in the telecommunications sector. When customers discontinue service, the company incurs immediate recurring revenue losses alongside unrecouped acquisition costs. The task is to create a predictive model to predict a customer will churn or retain their service based on several parameters.

1. IDENTIFYING THE PROBLEM STATEMENT

- The problem statement is to predict whether a customer will churn or retain their telecom services.

2. STAGES

- Machine Learning
- Supervised Learning
- Classification (Binary)

DATASET INFORMATION:

- The dataset has a total of 7043 rows and 21 columns
- There are 20 feature columns and 1 target column
- Target variable – Churn
- There are 13 binary columns with categorical data.
- Target Column (Churn) has a class imbalance where count of churn is lesser compared to retained customers count.

DATA PREPROCESSING AND ANALYSIS:

1. Data Pre Processing:

- Total Charges column had 11 whitespace strings which corresponds to new customers. So, they were imputed to 0 and also the data type was converted from object to float.
- Binary categorical columns were converted to numerical data using label encoding.
- Multi class categorical data was pre processed using One-Hot encoding.
- Customer ID column was dropped.

2. Exploratory Data Analysis:

- Firstly there were no outliers in the data.
- Customers on Month-to-month contracts have higher churn rate (42.7%) compared to those on One-year (11.3%) or Two-year (2.8%) contracts.

- Fiber optic service customers have higher churn rates as their monthly charges are higher.
- Tenure plays a major role on churn majorly customers churn within the first 5 months or their tenure.
- Customers utilizing Electronic check churn significantly more often than those using automated bank transfers or credit cards.

3. Imbalance handling, feature scaling and data splitting:

- **SMOTE** was applied on training set to balance the minority churn class in the data.
- **Standard Scaler** was applied on (tenure, total charges and monthly charges) columns as they had continuous data.
- The dataset was split into 80% training set and 20% testing set.

Feature selection and Model Training:

1. Model Training with all features:

- Classification models Logistic regression, SVM, Random forest, decision tree, KNN and naïve bayes algorithms were trained with all the 23 features using Gridsearch.

	Model	Accuracy	Precision	Recall	F1-Score	ROC-AUC	Best Params
0	Logistic Regression	0.7417	0.5039	0.6957	0.5845	0.8108	{'C': 10, 'penalty': 'l2', 'solver': 'saga'}
1	Naive Bayes	0.7353	0.4954	0.7391	0.5932	0.8106	{'var_smoothing': np.float64(0.000483293023857...
2	Random Forest	0.7630	0.5407	0.6141	0.5751	0.8068	{'class_weight': 'balanced', 'criterion': 'gin...
3	SVM	0.7260	0.4809	0.6141	0.5394	0.7766	{'C': 10, 'gamma': 'scale', 'kernel': 'rbf'}
4	KNN	0.6891	0.4345	0.6304	0.5144	0.7239	{'algorithm': 'ball_tree', 'n_neighbors': 3, '...
5	Decision Tree	0.7097	0.4549	0.5625	0.5030	0.6630	{'criterion': 'entropy', 'max_features': 'sqrt...

2. Model training with SelectK feature selection:

- All the 6 classification models were trained with selectk algorithm with different k values and with f_classif score function.
- **K = 15**

	Model	Accuracy	Precision	Recall	F1-Score	ROC-AUC	Best Params
0	Logistic Regression	0.7388	0.5000	0.7690	0.6060	0.8261	{'C': 1, 'penalty': 'l2', 'solver': 'lbfgs'}
1	Naive Bayes	0.7197	0.4772	0.7663	0.5881	0.8148	{'var_smoothing': np.float64(0.001438449888287...
2	Random Forest	0.7637	0.5434	0.5951	0.5681	0.8063	{'class_weight': 'balanced', 'criterion': 'gin...
3	SVM	0.7438	0.5068	0.7038	0.5893	0.7941	{'C': 10, 'gamma': 'scale', 'kernel': 'rbf'}
4	KNN	0.7076	0.4544	0.5951	0.5153	0.7265	{'algorithm': 'auto', 'n_neighbors': 3, 'weigh...
5	Decision Tree	0.7253	0.4773	0.5435	0.5083	0.6665	{'criterion': 'entropy', 'max_features': 'sqrt...

- **K = 7**

	Model	Accuracy	Precision	Recall	F1-Score	ROC-AUC	Best Params
0	Logistic Regression	0.7260	0.4844	0.7609	0.5920	0.8205	{'C': 1, 'penalty': 'l2', 'solver': 'saga'}
1	Naive Bayes	0.6657	0.4265	0.8125	0.5594	0.8051	{'var_smoothing': np.float64(0.112883789168468...
2	SVM	0.7189	0.4767	0.7799	0.5918	0.7952	{'C': 10, 'gamma': 'scale', 'kernel': 'rbf'}
3	Random Forest	0.7296	0.4846	0.5571	0.5183	0.7765	{'class_weight': 'balanced_subsample', 'criter...
4	KNN	0.7168	0.4689	0.6359	0.5398	0.7663	{'algorithm': 'auto', 'n_neighbors': 7, 'weigh...
5	Decision Tree	0.7133	0.4571	0.5217	0.4873	0.6534	{'criterion': 'gini', 'max_features': 'sqrt', ...

- **K = 5**

	Model	Accuracy	Precision	Recall	F1-Score	ROC-AUC	Best Params
0	Logistic Regression	0.7268	0.4853	0.7609	0.5926	0.8176	{'C': 1, 'penalty': 'l2', 'solver': 'lbfgs'}
1	Naive Bayes	0.6686	0.4304	0.8315	0.5672	0.8084	{'var_smoothing': np.float64(0.3359818286283782)}
2	Random Forest	0.7331	0.4925	0.7147	0.5831	0.8013	{'class_weight': 'balanced_subsample', 'criter...
3	Decision Tree	0.7395	0.5010	0.7120	0.5881	0.7975	{'criterion': 'entropy', 'max_features': 'sqrt...
4	SVM	0.7260	0.4846	0.7690	0.5945	0.7908	{'C': 10, 'gamma': 'scale', 'kernel': 'rbf'}
5	KNN	0.7480	0.5182	0.5027	0.5103	0.7698	{'algorithm': 'brute', 'n_neighbors': 7, 'weig...

3. Model training with PCA

- Again the same 6 classification models were trained with PCA with N features 4,10,15
- **N = 4**

	Model	Accuracy	Precision	Recall	F1-Score	ROC-AUC	Best Params
0	Logistic Regression	0.7509	0.5164	0.7255	0.6034	0.8106	{'C': 1, 'penalty': 'l2', 'solver': 'lbfgs'}
1	Naive Bayes	0.7353	0.4953	0.7201	0.5869	0.8022	{'var_smoothing': np.float64(0.004281332398719...
2	SVM	0.7367	0.4973	0.7527	0.5989	0.8010	{'C': 10, 'gamma': 'auto', 'kernel': 'rbf'}
3	Random Forest	0.7360	0.4955	0.5924	0.5396	0.7856	{'class_weight': 'balanced_subsample', 'criter...
4	KNN	0.7111	0.4600	0.6087	0.5240	0.7288	{'algorithm': 'ball_tree', 'n_neighbors': 3, '...
5	Decision Tree	0.6977	0.4353	0.5299	0.4779	0.6440	{'criterion': 'gini', 'max_features': 'sqrt', ...

- **N = 10**

	Model	Accuracy	Precision	Recall	F1-Score	ROC-AUC	Best Params
0	Logistic Regression	0.7445	0.5077	0.7174	0.5946	0.8112	{'C': 0.01, 'penalty': 'l2', 'solver': 'liblin...
1	Naive Bayes	0.7232	0.4804	0.7337	0.5806	0.7977	{'var_smoothing': np.float64(0.012742749857031...
2	SVM	0.7402	0.5019	0.7310	0.5951	0.7937	{'C': 10, 'gamma': 'scale', 'kernel': 'rbf'}
3	Random Forest	0.7615	0.5386	0.6060	0.5703	0.7935	{'class_weight': 'balanced_subsample', 'criter...
4	KNN	0.7076	0.4560	0.6196	0.5253	0.7296	{'algorithm': 'auto', 'n_neighbors': 3, 'weigh...
5	Decision Tree	0.6977	0.4319	0.5000	0.4635	0.6356	{'criterion': 'gini', 'max_features': 'sqrt', ...

- N = 15

	Model	Accuracy	Precision	Recall	F1-Score	ROC-AUC	Best Params
0	Logistic Regression	0.7544	0.5213	0.7310	0.6086	0.8121	{'C': 0.01, 'penalty': 'l2', 'solver': 'liblin...
1	Naive Bayes	0.7424	0.5047	0.7337	0.5980	0.7955	{'var_smoothing': np.float64(0.004281332398719...
2	Random Forest	0.7573	0.5337	0.5598	0.5464	0.7875	{'class_weight': 'balanced_subsample', 'criter...
3	SVM	0.7346	0.4938	0.6495	0.5610	0.7827	{'C': 10, 'gamma': 'scale', 'kernel': 'rbf'}
4	KNN	0.6877	0.4283	0.5842	0.4943	0.7130	{'algorithm': 'auto', 'n_neighbors': 3, 'weigh...
5	Decision Tree	0.6764	0.4060	0.5163	0.4545	0.6264	{'criterion': 'entropy', 'max_features': 'sqrt...

4. Model training with LDA

- All the classification models were trained with LDA and the metrics results with LDA = 1 were

	Model	Accuracy	Precision	Recall	F1-Score	ROC-AUC	Best Params
0	SVM	0.7509	0.5159	0.7473	0.6104	0.8146	{'C': 1, 'gamma': 'scale', 'kernel': 'linear'}
1	Naive Bayes	0.7459	0.5093	0.7473	0.6057	0.8146	{'var_smoothing': np.float64(0.001438449888287...
2	Logistic Regression	0.7516	0.5170	0.7446	0.6102	0.8146	{'C': 0.1, 'penalty': 'l2', 'solver': 'libline...
3	KNN	0.7161	0.4683	0.6413	0.5413	0.7572	{'algorithm': 'auto', 'n_neighbors': 7, 'weigh...
4	Random Forest	0.6969	0.4382	0.5679	0.4947	0.7095	{'class_weight': 'balanced', 'criterion': 'gin...
5	Decision Tree	0.6849	0.4224	0.5625	0.4825	0.6469	{'criterion': 'gini', 'max_features': 'sqrt', ...

CONCLUSION:

- Logistic regression has performed well when trained with SelectK with 7 features the model has the highest ROC-AUC and F1 score compared to other algorithms
- The naïve bayes algorithm has got the highest recall but its precision values are lower it causes reliability concerns.
- Other models has performed well on some metrics at the same time poor on some other metrics
- Therefore the Logistic regression model has overall balanced performance so it has been chosen as the best model.